

Summary & Objective: A curious and adaptive learner with strong domain expertise and outstanding communication skills, passionate about driving insights and building a meaningful and sustainable career in the U.S. data science industry.

EDUCATION

University of Michigan - Ann Arbor	Ann Arbor, MI
MS, Applied Statistics	08/2023 - 05/2025 (expected)
Virginia Polytechnic Institute and State University (Virginia Tech)	Blacksburg, VA
Double Major: B.S. Mathematics / B.S. Computer Modeling Data Analysis	08/2019 - 05/2023
Double Minor: BS Statistics / BE Computer Science	11/2021 - 05/2023

PROGRAMMING & QUALIFICATIONS

- Data Science:** PySpark, Hadoop, ETL, AWS, Machine Learning (XGBoost, Random Forest, CNNs, SVM, Logistic Regression, KNN), Statistical Analytics (A/B testing, EDA, Time Series Analysis), BI Dashboard (Tableau /Quarto/Power BI), NLP (NLTK)
- Programming Languages:** SQL, Python (TensorFlow, Scikit-learn, Pytorch), JAVA, C++, R, Linux, HTML/CSS
- Statistical Tools:** SAS (Certified Specialist), Excel, STATA, MATLAB

PROGRAMMING & QUALIFICATIONS

Penske Logistics	Reading, PA
Data Scientist Intern in Customer Experience Team	05/2024 - 07/2024
- Participated in designed and deploying the key components of customer analytics platform supported with big data (1MM+ transaction records) in SAS and Python by combing <i>RFM analysis</i> with <i>K-mean clustering</i> and ML-based churn prediction to fulfill business requirements of model centralization and improve accuracy of personalized marketing strategies (20% higher retention rate). - Validated platform enhancements and data inputs weekly by compiling performance reports in collaboration with senior analysts. - Teamed with senior scientists to automate a data monitoring process in Python for Penske data warehouse and develop a web-based visualization tool for business teams to identify and override/correct +15 types of data anomalies including inconsistent data types, outliers, improper data proxy usage, and feeding issues of upstream data sources. - Provided strategic deliveries and remediation plans for self-identified data issues or ad hoc business requirements. - Maintained the memory allocation for all Spark jobs and coordinated across end users for production releases, monitored the cluster/queue usage and run log of daily batches to ensure system stability.	

PROJECT EXPERIENCE

Petals to the Metal – Flower Classification	Kaggle Competition
- Implemented a CNN-based flower classification system in Python using TensorFlow by building an efficient TFRecord data pipeline to handle a 5GB+ image dataset, achieving a 91% accuracy (top 10 on Leaderboard). - Leveraged pretrained <i>CNNs</i> (<i>ResNet50</i>, <i>EfficientNetB0</i>) and optimized model performance through data augmentation and a combination of <i>GridSearch</i>, <i>Hyperband</i>, and manual tuning, boosting F1-score by 10% and reducing GPU inference time by 25%. - Authored clear documentation and maintained robust version control using Git and GitHub, ensuring team collaboration and reproducibility across development and competition submissions.	
Customer Segmentation Based on E-commerce Data	University of Michigan- Ann Arbor
- Built a scalable ETL pipeline using Spark and Hadoop to ingest, process, and transform over 10M+ records from five data sources (transaction, payment, location, etc.) for 146k customers, reducing pipeline run time by 50% through optimized Python-based processing with enhanced tree-structured data outputs given business request. - Conducted customer segmentation using <i>PCA</i> for dimensionality reduction, followed by <i>K-means</i> and <i>DBSCAN</i> clustering; generated 9 behavioral profiles with clear consumption behavior patterns and identified high-value users (top 5% contributed 60% of GMV). - Delivered interactive BI dashboards in Tableau to support real-time updates of segmentation results and customer behavior tracking, enabling data-driven targeted marketing strategies.	
Supervised Churn Prediction	Penske Logistics
- Developed and deployed churn prediction model utilizing supervised learning algorithms (<i>Logistic Regression</i>, <i>Random Forests</i>, and <i>XGBoost</i>) with approved model performance of 0.92 AUC-ROC score and reducing false positives by 18.6%. - Reduced dimensionality from 155 candidate variables using a two-step approach: applied SAS VARCLUS to group in 10+ clusters resulting in 35 variables; further refined by logistic regression model to select the most predictive 8 variables for final model training.	
Intelligent Course Assignment System – Capstone Project Allocation Engine	Virginia Tech
- Designed and implemented an intelligent allocation system based on <i>simulated annealing</i> in Python to dynamically match 200+ students with 30+ projects by analyzing interest survey data and quantifying project requirements. - Reverse-engineered faculty heuristics into 8 quantifiable constraints (e.g., interest alignment, skill complementarity, workload balance), enabling globally optimized assignment via temperature-controlled search ($T_0 = 1000 \rightarrow T_{min} = 1$). - Authored a 12-page technical white paper and created an animated visualization to communicate algorithm principles to non-technical stakeholders; delivered 5 milestone presentations to faculty panels, contributing to the platform’s adoption as the department’s standard allocation tool, used by 600+ students over 3 consecutive semesters.	